DeXposure-Claw: An Agentic System for DeFi Risk Monitoring and Decision Recommendation Based on a Graph Time-Series Foundation Model

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Abstract

Decentralised finance (DeFi) has become a densely interconnected financial system in which credit exposures propagate through tokenised liabilities across protocols, making timely *risk monitoring* and *decision recommendation* essential for investors, protocol operators, and regulators. Existing agentic pipelines for DeFi risk lack a dedicated forecasting backbone for graph-structured time series, which limits their data efficiency and uncertainty calibration. Building on DeXposure-FM, a graph time-series foundation model pre-trained on 43.7 million exposure-graph entries, we present two contributions. First, *DeXposure-Claw*: a four-layer query architecture—FM prediction, deterministic monitoring and scenario, LLM decision, and a data-health/confidence safety gate—that turns predicted exposure graphs into supervisory tickets. Second, *DeXposure-Bench*: a six-axis evaluation suite for FM + LLM decision pipelines on temporal exposure graphs, with a regulator-aligned ground truth defined by absolute exposure loss. The bench surfaces a non-trivial Pareto trade-off rather than uniform improvement: persistence remains strong on point-metric forecasting and ticket precision, while the FM turns forecasting into a deployable risk signal through trend consistency, calibrated uncertainty (PI coverage 0.912 against a 0.90 target), and robustness. In layer-wise decision ablations, the FM and LLM layers make additive contributions to ticket F1 ($+32\%$ from FM and $+27\%$ from LLM), while the safety gate improves explanation quality at a small F1 cost. We release the harness, dataset, and reference implementations to support extension.

Index Terms

Agentic system, graph, time series, foundation model, decentralised finance, financial network, risk modelling.

1 Introduction

In May 2022, the Terra/Luna collapse erased over \$40 billion in value within 48 hours, triggering cascading liquidations across lending protocols, DEXs, and yield aggregators that held UST-denominated assets. Five months later, the FTX insolvency exposed hidden cross-protocol exposures and wiped out another \$8 billion. These crises share a common pattern: shocks propagated through an interconnected web of tokenised liabilities—credit exposures—that no single participant fully observed. In decentralised finance, exposures are high-dimensional, time-varying, and often implicit; data are noisy and incomplete; and crises unfold within days, leaving limited time for manual assessment.

Automated *agentic* pipelines offer a principled response: an agent can ingest on-chain measurements, maintain a belief over the evolving exposure network, surface early-warning signals, stress-test plausible shocks, and recommend risk-mitigating actions with auditable rationale. However, existing agentic proposals for DeFi risk largely rely on general-purpose large language models or static heuristics, and lack a dedicated forecasting backbone for graph-structured time series. Without accurate and data-efficient prediction of network dynamics, monitoring agents suffer from high false-alarm rates and delayed detection, while decision agents risk recommending actions based on brittle or poorly calibrated beliefs. In high-stakes contexts, uncertainty calibration and robustness to missingness and regime shifts are not optional—they determine whether an agent is trusted and deployable.

This paper addresses that bottleneck by building an agentic system around a strong graph time-series foundation model, and—in tandem—by providing the evaluation suite the community needs to compare such systems. We leverage DeXposure-FM, a foundation model trained on over 43.7 million data entries from DeFi exposure networks, designed to forecast the evolution of credit-exposure graphs across multiple horizons. On top of this backbone we present DeXposure-Claw, a modular four-layer query architecture that follows a *forecast–measure–decide* paradigm, and DeXposure-Bench, a standardised evaluation suite that benchmarks FM+LLM decision pipelines on a shared regulator-aligned ground truth.

Our contributions are:

1) **DeXposure-Claw: a four-layer query architecture for DeFi exposure-network risk** (§4). We design and implement an end-to-end system organised as four layers—FM prediction, deterministic monitoring and scenario engines, LLM-based decision, and a safety gate—that transforms forecasted exposure graphs into auditable supervisory tickets. Every

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component is deterministic and reproducible: five network risk metrics with rolling baselines, five standardised stress scenarios with CVaR aggregation, and a four-level playbook governed by data-health and confidence gates. While prior agentic work in DeFi targets transaction verification [1] or intent mining [2], and graph-based foundation models for risk have been explored only in traditional finance [3], no existing system unifies these capabilities for DeFi exposure networks.

2) **DeXposure-Bench: an evaluation suite for FM+LLM decision pipelines** (§5). We release the first standardised harness that scores a candidate pipeline on six axes—forecasting quality, streaming anomaly detection, predictive uncertainty, what-if scenario fidelity, supervisory decision quality, and data-quality robustness—on the 16,610-protocol DeXposure dataset. The suite introduces a regulator-aligned ground truth defined by *absolute* weight loss rather than fractional change, which restores discrimination between methods that target central versus peripheral nodes. Generic LLM-agent benchmarks (SWE-bench, AgentBench, HELM) do not evaluate domain-grounded decisions, and temporal-graph benchmarks (TGB, OGB) stop at structural prediction; DeXposure-Bench fills the niche.
3) **Deployable FM signals and agentic layer-wise gains** (§6). On DeXposure-Bench we show that persistence remains a strong static-rank baseline, but the FM turns forecasting into a deployable monitoring signal: TrendCons@h4 reaches 0.628 while persistence is pinned at 0, relative degradation is lower (0.113 vs 0.148), and prediction intervals are calibrated (coverage 0.912 vs the 0.90 target). Downstream, the FM prediction layer and LLM decision layer each contribute independently to ticket F1 (+32% and +27% respectively in controlled ablations), while the safety gate raises LLM-as-judge explanation quality from 2.48 to 2.69 / 5. Components that appear inactive under clean test conditions (data-health gating, multi-horizon diversity) are shown to be safety reserves that activate under data degradation and crisis dynamics.

We will release the implementation and benchmark artifacts with the project repository.

2 Related Works

This section positions DeXposure-Claw against five neighbouring threads: financial-network contagion, temporal graph learning, foundation models for time series and graphs, LLM agents in finance and DeFi, and benchmark design.

2.1 Financial Network Risk and DeFi Contagion

Systemic-risk analysis in traditional finance has long treated the financial system as a network of balance-sheet obligations, funding links, and correlated asset exposures. Clearing-network models formalise how defaults propagate through interbank liabilities [4], while subsequent work studies when dense connections stabilise the system and when they amplify shocks [5], [6]. DebtRank-style measures further emphasise that a node can be “too central to fail” even when it has not yet defaulted [7]. These models motivate our focus on network position, concentration, and stress-test loss rather than protocol-level indicators alone.

DeFi changes the object being measured. Smart contracts replace part of the legal and operational layer, but composability, shared collateral, stablecoins, bridges, and liquidations create new contagion channels [8], [9], [10]. Recent work measures DeFi risk from aggregate deposits and borrowings [11], maps microscopic and systemic mechanisms across TradFi and DeFi [12], and studies fragility in TVL dynamics [13]. In contrast, our setting is an inter-protocol credit-exposure network: the units of analysis are weighted links between protocols, and the downstream task is not only to detect stress but also to produce traceable supervisory tickets.

2.2 Temporal Graph Learning for Dynamic Risk

Graph neural networks have become a natural tool for financial networks because they use both node attributes and relational structure. Prior work shows that GNNs can improve systemic-importance classification and systemic risk computation in simulated or interbank-style financial networks [14], [15], building on general graph convolution and attention architectures [16], [17]. These studies support the premise that risk is partly encoded in graph topology, not just in independent node features.

Our problem is additionally temporal: exposure weights, active protocols, and contagion paths evolve week by week. Dynamic-graph models such as DySAT, TGAT, EvolveGCN, and Temporal Graph Networks model changing neighbourhoods, temporal attention, recurrent graph filters, or event streams [18], [19], [20], [21]. Temporal graph benchmarks standardise link-prediction and node-classification tasks on dynamic graphs [22]. However, these settings typically optimise task-specific structural prediction. DeXposure-Claw instead uses temporal graph forecasts as inputs to monitoring, scenario stress tests, and LLM-based decisions; the forecasting layer is therefore evaluated by its effect on calibrated risk signals and recommendation tickets, not only by edge-level accuracy.

2.3 Foundation Models for Time Series and Graphs

Foundation models aim to replace narrowly trained task models with reusable representations learned from broad pre-training data [23]. Earlier neural forecasting architectures such as Temporal Fusion Transformers, Informer, and Autoformer established multi-horizon attention-based forecasting for long and heterogeneous time series [24], [25], [26]. More recently, Chronos, Lag-Llama, and TimesFM show that large pretrained sequence models can transfer across forecasting tasks and provide probabilistic forecasts [27], [28], [29]. In finance and economics, LLMs have also been used to extract signals from news and policy text for nowcasting pipelines [30], [31], [32].

Graph foundation modelling is less settled. Graphormer demonstrates that transformers can be made graph-aware through structural encodings [33]; self-supervised graph representation learning methods such as Deep Graph Infomax, GraphCL, and GraphMAE pretrain encoders through mutual information, contrastive augmentations, or masked attribute reconstruction [34], [35], [36]; and GraphPFN and TabPFN-style work explores how pretrained tabular predictors can transfer to graph-structured tasks [37], [38]. Heterogeneous graph pretraining has also been applied to default-risk propagation among bond issuers [3]. These systems motivate a foundation-model backbone for risk forecasting, but they do not directly address weighted temporal DeFi exposure graphs or the downstream requirements of uncertainty-gated supervisory decision making.

2.4 LLM Agents in Finance and DeFi

LLM agents are increasingly used for financial analysis, trading, and blockchain security. General agent patterns such as ReAct combine reasoning traces with tool-using actions [39], while financial LLM work such as FinGPT adapts language models to financial data and tasks [40]. In crypto portfolio management, multi-agent LLM systems coordinate market analysis and allocation decisions [41], and BlindTrade combines graph encoders with LLM agents for anonymised stock portfolio optimisation [42]. These systems are closest to ours in combining graph representations with language-model reasoning, but their objective is portfolio construction rather than systemic-risk supervision.

Within DeFi, agentic systems have focused mainly on security auditing, transaction understanding, and manipulation detection. Arbiter uses Graph-of-Thoughts reasoning for intent–transaction alignment [1]; Know Your Intent mines transaction intent with a multi-perspective LLM agent [2]; Heimdallr and Knowdit target smart-contract vulnerability detection [43], [44]; and PMDetector applies LLM reasoning to price manipulation detection [45]. Watson et al. combine graph anomaly detection with LLM-generated explanations for Bitcoin fraud [46]. These works show that LLMs can help explain or audit crypto systems, but they do not couple the agent with a dedicated graph time-series forecaster for network-level exposure risk.

2.5 Benchmarks for Graph, Forecasting, and Agentic Systems

The benchmark literature provides useful pieces but not the complete evaluation target required here. OGB and TGB standardise graph and temporal graph evaluation [22], [47]; time-series forecasting work emphasises temporally valid splits and robust model selection [48]; and general LLM-agent benchmarks such as HELM, SWE-bench, and AgentBench evaluate language capabilities, software repair, or generic agent behaviour [49], [50], [51].

DEXPOSURE-BENCH occupies a different point in this space. It evaluates forecasting, early warning, calibration, scenario stress fidelity, decision quality, and data-quality robustness in one harness, with ground truth tied to absolute exposure loss. This lets us ask whether a foundation-model forecast actually improves downstream risk monitoring and ticket generation, and also whether safety gates reduce harmful recommendations under missingness or weak evidence.

3 NOTATION AND PRELIMINARIES

This section introduces the key terminology and notation used throughout the paper.

3.1 Credit Exposure in DeFi Networks

Let $X_G(q)$ be the set of tokens issued (or generated) by protocol q, and let $X_p(t)$ be the set of tokens held (or locked) by protocol p at time t. We say that protocol p has *credit exposure* to protocol q at time t if p holds at least one token issued by q:

$$X_p(t) \cap X_G(q) \neq \varnothing. \tag{1}$$

Equivalently, p is exposed to liabilities of q: if tokens issued by q devalue or become illiquid, p incurs losses. This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi, where exposures emerge through tokenised claims rather than explicit bilateral contracts. In practice, these dynamics can be tracked by reconstructing protocol-level TVL time series and monitoring how token composition changes over time [52].

Figure 1 shows how a single workflow can generate layered exposures. A user deposits ETH into Lido and receives stETH; stETH is wrapped into wstETH; wstETH is then deposited into Pendle, which splits it into PT-wstETH and YT-wstETH. On balance sheets, each step turns the upstream token into the downstream protocol's asset, so exposures stack: Pendle is exposed to the wstETH wrapper, which is exposed to Lido. If Lido suffers slashing or stETH de-pegs from ETH, losses can propagate through wstETH into Pendle and its PT/YT holders. This illustrates why DeFi exposures are naturally multi-layered and why DeXposure aims to measure them.

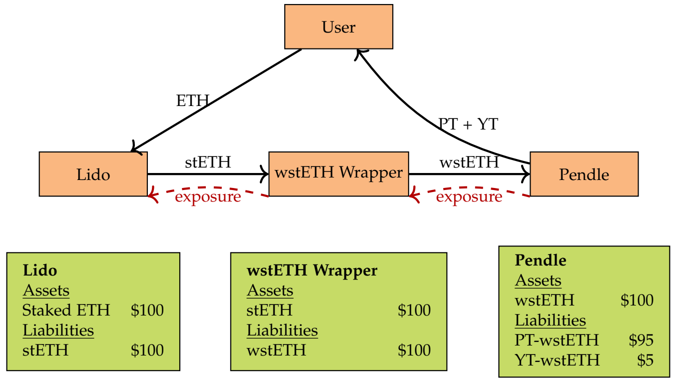

Fig. 1. Multi-layer credit exposure in DeFi. A user stakes ETH with Lido, receiving stETH, which is wrapped into wstETH and deposited into Pendle. Each protocol holds the liability of the preceding one, creating a chain of credit exposures from Pendle to Lido.

3.2 Mapping Tokens to Issuing Protocols

To implement Eq. (1), we must identify the issuing (or managing) protocol for each token. Let $\mathcal{P}$ be the set of protocols and $\mathcal{C}$ the set of blockchains. For any token σ appearing in protocol p, a *mapping function* $M(\sigma)$ returns the protocol q that issues or governs σ. We construct M via the following fallback pipeline:

1) *Metadata lookup:* match tokens to issuing protocols using DefiLlama metadata when available [53].
2) *Manual mapping:* for high-TVL tokens without metadata links, assign issuers using documentation and expert judgment.
3) *Text-vector similarity:* otherwise, vectorise token and protocol descriptions using TF-IDF [54] and assign $M(\sigma)$ as the most similar protocol, provided the cosine similarity exceeds a threshold θ.
4) *Primary market tokens:* if no match is obtained, treat the token as its own protocol (e.g., WETH).

This procedure attributes token movements to the appropriate issuer even when tokens are bridged or wrapped, and is used in curating DeXposure.

3.3 Network Representation and Value Flows

Over a discrete interval $\tau = t_2 - t_1$, we represent exposures by a weighted directed graph $G_\tau = (P_\tau, E_\tau)$, where P_τ is the set of active protocols and E_τ the directed edge set. Each node $p \in P_\tau$ receives a *node weight* equal to the value of its held assets at t_2:

$$w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2,\sigma}, \tag{2}$$

where $v_{t_2,\sigma}$ is the USD value of token σ at time t_2. Nodes below a threshold θ may be pruned.

To define the *edge weight* from p to q, we first compute token-level value flows:

$$F^\sigma_{pq,\tau} = \begin{cases} \max(0,\, -\Delta S^\sigma_{p,\tau}), & \text{if } \Delta S^\sigma_{p,\tau} < 0, \\ \max(0,\, \Delta S^\sigma_{q,\tau}), & \text{if } \Delta S^\sigma_{q,\tau} \geq 0, \end{cases} \tag{3}$$

where $\Delta S^\sigma_{p,\tau} = v^\sigma_{p,t_2} - v^\sigma_{p,t_1}$ is the change in the USD value of token σ held by p over τ. Intuitively, if p reduces holdings while q increases holdings, σ is interpreted as flowing from p to q; the construction enforces non-negativity.

The *edge weight* $w_\tau(e_{pq})$ aggregates flows over tokens mapped to issuer q:

$$w_\tau(e_{pq}) = \sum_{\sigma \in X_{pq,\tau}} F^\sigma_{pq,\tau}, \tag{4}$$

where $X_{pq,\tau}$ is the set of tokens issued by q that move from p to q over τ. Positive $w_\tau(e_{pq})$ corresponds to increasing exposure from p to q; for single-token protocols, that token's flow equals the edge weight. In summary, we compute G_τ by aggregating holdings at t_1 and t_2, assigning node weights, pruning small nodes, and summing positive token flows into issuer-specific edge weights. The resulting graph sequence captures the evolving credit-dependency structure curated in DeXposure.

4 Algorithm

This section specifies the agentic system built on DeXposure-FM for two tasks: network risk monitoring (Section 4.3–4.4) and decision recommendation (Section 4.5).

Figure 2 provides an architectural overview of the full system.

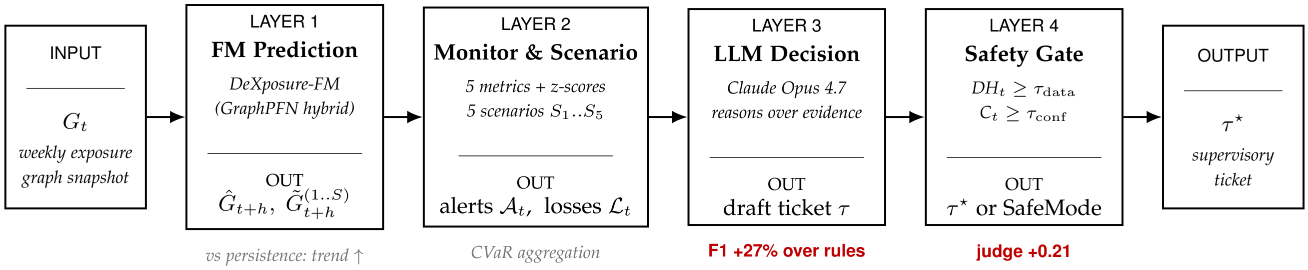

Fig. 2. DEXPOSURE-CLAW: a four-layer query architecture. The system ingests the weekly exposure graph G_t, runs FM prediction (Layer 1), monitoring and scenario engines (Layer 2), an LLM decision layer (Layer 3), and a safety gate (Layer 4) that emits either the supervisory ticket $\tau^\star$ or a safe-mode notice.

ID	Metric	Definition
N1	Systemic Importance	$\max_{p \in V} \text{PageRank}(\widehat{G}_{t+h})_p$
N2	HHI Concentration	$\sum_p \big(d_p / \sum_q d_q\big)^2$, $d_p =$ weighted out-degree
N3	Network Density	$\lvert E_{t+h}\rvert \,/\, \lvert V\rvert(\lvert V\rvert - 1)$
N4	PageRank Gini	$\text{Gini}\big(\{\text{PR}_p\}_{p\in V}\big)$
N5	Degree Gini	$\text{Gini}\big(\{d_p\}_{p\in V}\big)$

TABLE 1
Monitoring functionals Φ. PageRank uses power iteration with damping 0.85 and 10 iterations. Gini coefficient: $1 - 2\sum_{i=1}^{n}\sum_{k=1}^{i} x_{(k)} \,/\, (n \cdot \sum_k x_{(k)})$, where $x_{(k)}$ are sorted values.

4.1 System Interface

At each decision epoch t, the agent observes a credit-exposure snapshot $G_t = (V, E_t, W_t)$ and node covariates X_t, and interacts with DEXPOSURE-FM through a single forecast primitive:

$$\mathcal{P}_{t,h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h), \quad \forall h \in \mathcal{H}, \tag{5}$$

where $\mathcal{P}_{t,h}$ denotes the model's predictive distribution over the future network G_{t+h}, and $\mathcal{H} = \{1, 4, 8, 12\}$ (weeks). From $\mathcal{P}_{t,h}$ we construct a representative predicted graph $\widehat{G}_{t+h}$ using a thresholded expected-weight operator:

$$\widehat{w}_{pq,t+h} := \Pr(e_{pq,t+h} = 1) \cdot \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1], \tag{6}$$

$$\widehat{E}_{t+h} := \{(p,q) : \Pr(e_{pq,t+h} = 1) > \pi_{\min}\}. \tag{7}$$

where $\pi_{\min}$ (default 0.2, tuned on validation) balances edge coverage and sparsity. For uncertainty quantification, we draw M Monte Carlo samples $\{\widetilde{G}^{(m)}_{t+h}\}_{m=1}^{M}$ by applying Bernoulli edge sampling and Gaussian weight perturbation to $\mathcal{P}_{t,h}$ (default $M = 50$).

The agent returns three outputs: (a) a set of monitoring alerts with evidence and confidence, (b) a scenario stress summary, and (c) a ranked list of recommendation tickets.

4.2 Data-Health Gating

Real-world DeFi graphs can be sparse, stale, or corrupted. Before producing actionable outputs, the agent computes a scalar data-health score from four deterministic checks—freshness, feature missingness, topology sanity (edge-to-node ratio), and discontinuity detection:

$$\begin{aligned} DH_t &:= \text{mean}\big(\text{fresh}_t,\ \text{miss}_t,\ \text{topo}_t,\ \text{disc}_t\big) \in [0,1], \\ \text{SAFE\_MODE}_t &:= \mathbb{I}\{DH_t < \tau_{\text{data}}\}. \end{aligned} \tag{8}$$

When DH_t falls below τ_{data} (default 0.7), the system enters safe mode: monitoring alerts are still emitted, but intervention-type recommendations are suppressed (Section 4.5).

4.3 Task I: Network Risk Monitoring

Monitoring metrics (Φ).: Risk in exposure networks manifests as concentration (a few protocols dominate), fragility (shocks propagate widely), and instability (structure shifts rapidly). To capture these dimensions, the monitoring agent evaluates five functionals on $\widehat{G}_{t+h}$:
Formally, at each horizon h we compute:

$$\mathbf{N}_{t,h} := \Phi(\widehat{G}_{t+h}, X_t) \in \mathbb{R}^5, \qquad \widetilde{\mathbf{N}}^{(m)}_{t,h} := \Phi(\widetilde{G}^{(m)}_{t+h}, X_t). \tag{9}$$

ID	Scenario	Target	Shock
S1	Single protocol failure	Highest-weight node	100%
S2	Bridge cluster failure	All bridge-category nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector-wide shock	All lending nodes	30%
S5	Correlated stress	Top-10 nodes by weight	20%

TABLE 2
Standardised stress-scenario library $\mathcal{S}_0$.

Alert generation.: An alert fires when a predicted metric departs from its rolling historical baseline by more than z standard deviations (default $z = 2$, baseline window $L = 26$ weeks):

$$\mathcal{A}_t := \Big\{ a = (h, j) \ : \ \frac{|N_{t,h}^{(j)} - \mu_t^{(j)}|}{\max(\sigma_t^{(j)},\, \epsilon)} > z \Big\}, \tag{10}$$

where $(\mu_t^{(j)}, \sigma_t^{(j)})$ are the rolling mean and standard deviation of metric j over the last L observed snapshots, and $\epsilon = 10^{-8}$ prevents division by zero.

Evidence and attribution.: Each alert $a \in \mathcal{A}_t$ carries an evidence bundle $\mathrm{Ev}_t(a) = (N_{t,h}^{(j)},\, \mu_t^{(j)},\, \sigma_t^{(j)},\, \mathrm{Attr}_t(a),\, h)$, where $\mathrm{Attr}_t(a)$ is a ranked list of top-K contributing nodes/edges obtained via marginal contribution approximation: $\mathrm{Contrib}(e) := \Phi^{(j)}(\widehat{G}_{t+h}) - \Phi^{(j)}(\widehat{G}_{t+h} \setminus e)$ (default $K = 10$).

Uncertainty-aware confidence.: Alert reliability depends on three factors: data quality, forecast dispersion, and horizon length. Using Monte Carlo samples, we compute the coefficient of variation $\mathrm{CV}_{t,h}^{(j)} := \sigma(\{\widetilde{N}_{t,h}^{(m,j)}\}_m)/|\bar{N}_{t,h}^{(j)}|$ (clamped to $[0, 1]$) and define:

$$C_t(a) \ := \ DH_t \cdot \frac{1}{1 + \mathrm{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \ \in [0, 1]. \tag{11}$$

This score monotonically decreases with lower data health, wider predictive spread, and longer horizons, ensuring that unreliable alerts are automatically downweighted.

4.4 Scenario Engine

To quantify tail risk beyond metric-level alerts, the agent evaluates a standardised stress-scenario library $\mathcal{S}_0$. Each scenario applies a deterministic shock—removing or reducing edge weights of targeted nodes—and propagates losses through the network:

For each scenario s and horizon h, we compute the systemic loss on both the predicted graph and each MC sample:

$$L_{t,h}(s) := \frac{\sum_e w_e(\widehat{G}_{t+h}) - \sum_e w_e(\widehat{G}_{t+h}^s)}{\sum_e w_e(\widehat{G}_{t+h})}, \tag{12}$$

$$\widetilde{L}_{t,h}^{(m)}(s) := \ell\Big(\widetilde{G}_{t+h}^{(m)};\, s\Big). \tag{13}$$

where $\widehat{G}_{t+h}^s$ denotes the network after applying shock s. The scenario engine also tracks the number of *distressed* nodes (those losing $> 50\%$ of total edge weight) and a propagation depth estimate $\lfloor \log_2(\text{affected nodes} + 1) \rfloor + 1$.

CVaR aggregation.: To capture tail risk, we compute an empirical CVaR from the MC sample losses. Let $\lambda \in [0, 1]$ (default 0.5) control the tail fraction. We sort the M sample losses in descending order and average the worst $\lceil \lambda M \rceil$:

$$\mathrm{CVaR}_\lambda(s, h) := \frac{1}{\lceil \lambda M \rceil} \sum_{i=1}^{\lceil \lambda M \rceil} \widetilde{L}_{t,h}^{(i)}(s), \tag{14}$$

where $\widetilde{L}^{(1)} \geq \widetilde{L}^{(2)} \geq \cdots$ are order statistics. The scenario summary $\mathcal{R}_t$ ranks scenarios by $L_{t,h}(s)$ and retains the worst-case scenario across all horizons.

4.5 Task II: Decision Agent

Action space.: The decision module maps alerts and scenario results to recommendation tickets using a conservative playbook with four action types, ordered by escalation severity:

Feasibility constraints.: An intervention-type action u (those requiring safe-mode off) is feasible only when both the data-health gate and the confidence gate are satisfied:

$$u \in \mathcal{C}_t \quad \Longleftrightarrow \quad (\mathrm{SAFE\_MODE}_t = 0) \ \wedge \ \Big(\bar{C}_t \geq \tau_{\mathrm{conf}}\Big), \tag{15}$$

where $\bar{C}_t = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} C_t(a)$ is the mean alert confidence and τ_{conf} (default 0.6) gates interventions. Non-interventional actions (`Monitor`, `Investigate`) are always feasible when at least one alert exists.

Action	Severity	Weight w_u	Safe-mode OK?	Min. alerts
`Monitor`	Low	0.25	Yes	1
`Investigate`	Medium	0.50	Yes	1
`Recommend -Reduce`	High	0.75	No	2
`Contingency`	Critical	1.00	No	3

TABLE 3
Playbook action space $\mathcal{U}$ with severity weights and feasibility constraints.

Algorithm 1 DeXposure-Claw: one epoch

Require: State (G_t, X_t); horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\min})$; MC count M
Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$

$DH_t \leftarrow \text{DataHealth}(G_t, X_t);\ \text{SAFE\_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$
for all $h \in \mathcal{H}$ **do**
  $\mathcal{P}_{t,h} \leftarrow \text{DeXposure-FM.Forecast}(G_t, X_t, h)$
  $\widehat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h};\ \pi_{\min})$ ▷ Eq. (6)
  $\{\widetilde{G}^{(m)}_{t+h}\}_{m=1}^{M} \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$ ▷ Bernoulli edges + Gaussian weights
  $\mathbf{N}_{t,h} \leftarrow \Phi(\widehat{G}_{t+h}, X_t);\ \widetilde{\mathbf{N}}^{(m)}_{t,h} \leftarrow \Phi(\widetilde{G}^{(m)}_{t+h}, X_t)$ ▷ Table 1: N1..N5
  $(\mu^{(j)}_t, \sigma^{(j)}_t) \leftarrow \text{RollingStats}(\{\mathbf{N}_{t-l,h}\}_{l=1}^{L})$ ▷ $L = 26$ weeks
  $\mathcal{A}_{t,h} \leftarrow \{(h, j) : |N^{(j)}_{t,h} - \mu^{(j)}_t| / \max(\sigma^{(j)}_t, \epsilon) > z\}$ ▷ Eq. (10)
  Attach $\text{Ev}_t(a)$ and $C_t(a)$ for all $a \in \mathcal{A}_{t,h}$ ▷ Eq. (11), uses $\widetilde{\mathbf{N}}^{(m)}_{t,h}$ for CV
  for all $s \in \mathcal{S}_0$ **do**
    $L_{t,h}(s) \leftarrow \text{StressLoss}(\widehat{G}_{t+h}, s);\ \widetilde{L}^{(m)}_{t,h}(s) \leftarrow \text{StressLoss}(\widetilde{G}^{(m)}_{t+h}, s)$
    $\text{CVaR}_\lambda(s, h) \leftarrow \text{TailMean}(\{\widetilde{L}^{(m)}_{t,h}(s)\}_{m=1}^{M}, \lambda)$ ▷ Eq. (14)
  end for
end for
$\mathcal{A}_t \leftarrow \bigcup_{h \in \mathcal{H}} \mathcal{A}_{t,h}$
$\mathcal{R}_t \leftarrow \text{RankScenarios}(\{L_{t,h}(s), \text{CVaR}_\lambda(s, h)\}_{h,s})$
$\mathcal{D}_t \leftarrow \text{PlaybookDecision}($
  $\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE\_MODE}_t, \tau_{\text{conf}})$ ▷ Table 3
return $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

Ticket scoring.: Each feasible recommendation is scored by a multiplicative heuristic combining playbook severity, alert confidence, and scenario impact:

$$\text{Score}_t(u) := w_u \cdot \bar{C}_t \cdot \text{Imp}_t, \tag{16}$$

where w_u is the severity weight (Table 3), $\bar{C}_t$ is the mean alert confidence, and $\text{Imp}_t := \frac{1}{|\mathcal{S}_0|} \sum_{s \in \mathcal{S}_0} \text{CVaR}_\lambda(s)$ aggregates scenario impact as the mean CVaR across all scenarios. When no scenarios are available, Imp_t defaults to 1.0 (neutral multiplier), ensuring alert-only tickets receive non-zero scores. The agent returns the top-N tickets (default $N = 10$) ranked by $\text{Score}_t(u)$.

Target selection.: For intervention tickets, targets are selected by collecting the top-K attributed nodes from all triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

4.6 End-to-End Procedure

Algorithm 1 summarises the full agent loop.

5 DeXposure-Bench: An Evaluation Suite for FM+LLM Decision Pipelines

5.1 Motivation: an evaluation gap

Foundation-model pipelines that pair a domain forecaster with a large language model decision layer are proliferating across finance, supply-chain risk, and infrastructure monitoring. Despite this, the community lacks a standardised suite for evaluating the full *pipeline*—most existing benchmarks evaluate a single stage in isolation. Generic LLM-agent benchmarks (SWE-bench [50], AgentBench [51], HELM [49]) cover open-ended coding or tool use but not domain-grounded supervisory decisions. Temporal-graph benchmarks (TGB [22], OGB [47]) measure structural prediction quality but stop short of any decision metric. Domain-specific evaluations of DeFi risk forecasting exist as one-off experiments accompanying single papers, none of them packaged as a reusable harness.

DeXposure-Bench fills this gap with a single, reproducible suite that scores a candidate pipeline on six axes spanning forecast quality, anomaly detection, predictive uncertainty, what-if query fidelity, downstream decision quality, and data-quality robustness, on a shared 16,610-protocol DeFi exposure dataset with a *regulator-aligned* ground truth.

TABLE 4
DeXposure-Bench schema. Each row is an independently reportable benchmark; the full leaderboard requires all six.

ID	Capability tested	Primary metrics	Applicable methods
`b1_forecast`	Temporal graph prediction quality	PageRank MAE, HHI MAE, Gini MAE, trend consistency, rank correlation; per horizon $h \in \{1, 4, 8, 12\}$	Any predictor
`b2_warning`	Streaming anomaly detection	Precision, recall, F1, lead time, stability; per event $\times$ alert budget	Any monitor + predictor
`b3_calibration`	Predictive uncertainty quality	PI coverage @ 0.90 target, PI width, ECE, CRPS	Predictors with intervals
`b4_stress`	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@k; per scenario $S_1 \ldots S_5$	Predictor + scenario engine
`b5_decision`	Supervisory ticket quality	Precision, recall@k, F1, target stability, judge score, FIR	Predictor + decision policy
`b6_robustness`	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation	Any predictor

TABLE 5
DeXposure-Bench canonical split. All reported numbers in this paper come from the test split.

Split	Period	Weeks	Usage
Train	2020-03–2024-06	$\sim$220	Predictor training
Validation	2024-07–2024-12	$\sim$26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	$\sim$33	Frozen leaderboard

5.2 Suite schema

The suite is organised as six benchmarks, each producing an independently reportable scalar (or vector of scalars). Pipelines may report against any subset; the full leaderboard requires all six. Table 4 summarises each axis.

The six axes were chosen so that they are: (i) *non-overlapping* in what they measure—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework* (§4), so that ablations on the predictor layer (b1, b3, b6), monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

5.3 Dataset and split

The benchmark draws on the DeXposure weekly exposure-graph dataset [52], a directed weighted graph $G_t = (V_t, E_t, w_t)$ at each weekly timestep t, with V_t a subset of the 16,610-protocol universe observed over 2020-03–2025-08 (283 weeks). Node features carry log-TVL, number of token types, maximum token share, Shannon entropy of the token distribution, and a 15-class category one-hot. Edge weights are log-scaled USD inter-protocol exposures. We adopt the strict no-leakage temporal split of Table 5.

5.4 Ground truth: regulator-aligned absolute loss

Prior systemic-risk evaluations rank protocols by *fractional* weight change, which produces a ground-truth pool dominated by small protocols whose deterioration carries low systemic significance. This choice silently biases every downstream metric and is not what supervisory authorities monitor in practice.

DeXposure-Bench therefore defines its ground truth in terms of *absolute* exposure loss. Let $w_t(v) = \sum_{e \ni v} w_t(e)$ be the total weight incident on protocol v at time t. For horizon h and lookahead window W we define the *stressed set* $\mathcal{S}_t^h$ as the top π fraction of active protocols ranked by absolute loss:

$$\Delta_t^h(v) \;=\; w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h \;=\; \mathrm{top}_\pi\{v : w_t(v) > 0, \Delta_t^h(v) > 0\}. \tag{17}$$

We fix $\pi = 0.05$ throughout (yielding an average pool of $|\mathcal{S}_t^h| \approx 283$ protocols per week, stable across the test split) and the lookahead window $W = 4$ weeks.

Equation 17 is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline. Aligning the metric with absolute economic magnitude—rather than fractional change—restores discrimination between methods that target central versus peripheral nodes and matches how a supervisor would prioritise.

5.5 Methods supported out of the box

Nine reference methods are bundled with the harness, sufficient to populate a complete leaderboard from a fresh checkout. Each method exposes the same `predict_graph` or `decide_actions` interface, so adding a tenth method is a single-file change.

The seven listed in Table 6 together expose all four cuts of the framework ablation: (i) adding the predictor layer (`m1`→`m5`), (ii) swapping the decision layer (`m5`→`m6`), (iii) adding the safety gate (`m6`→`m7`), and (iv) removing the predictor from the LLM stack (`m2`↔`m6`).

TABLE 6
Reference methods bundled with DEXPOSURE-BENCH. Heuristic identifiers use the `h` namespace; learned methods use `m`.

ID	Predictor	Policy	Role
`h1_weighted_degree`	—	heuristic alert	Streaming anomaly baseline
`m1_persistence_rules`	$G_{t+h} = G_t$	rule engine	Decision baseline
`m2_snapshot_llm`	current snap	LLM	LLM-only ablation
`m3_evolvegcn`	EvolveGCN	—	Learned-GNN baseline
`m4_fm_only`	FM	—	FM-only ablation
`m5_fm_rules`	FM	rule engine	FM + rules
`m6_fm_llm`	FM	LLM	Full FM + LLM stack
`m7_fm_llm_gated`	FM	LLM + safety gate	Safety-gated FM + LLM

TABLE 7
DEXPOSURE-BENCH versus closest existing suites. A check mark indicates a covered axis; — indicates the axis is out of scope.

Suite	Pred.	Anom.	Calib.	Scen.	Decision
OGB [47]	✓	—	—	—	—
TGB [22]	✓	—	—	—	—
HELM [49]	—	—	—	—	✓
SWE-BENCH [50]	—	—	—	—	✓
AGENTBENCH [51]	—	—	—	—	✓
DEXPOSURE-BENCH (this work)	✓	✓	✓	✓	✓

5.6 Reporting protocol

To enter the leaderboard, a submission produces one JSON file per {benchmark, method} pair following the harness's published schema, and a single `results.json` aggregating the suite. Required reports for the headline table are b1, b3, b5, and b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. All metrics are computed on the frozen test split of Table 5; training and hyperparameter tuning may use train + validation freely. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files so that any reported number can be reproduced bit-exactly.

5.7 Positioning in the benchmark landscape

Table 7 compares DEXPOSURE-BENCH against the closest existing evaluation suites. No prior suite covers a domain-grounded decision metric on top of a forecasting backbone, which is the niche DEXPOSURE-BENCH addresses.

5.8 Reproducibility and extensibility

The harness, dataset, and reference implementations are released under an open licence at the project repository. A complete leaderboard run, including all nine reference methods and the LLM evaluation pipeline, completes in under four GPU-hours on a single RTX 4090 plus approximately ten USD of LLM API spend, making the benchmark accessible to academic groups without industrial compute. Extending the suite—adding a method, an additional ground-truth formulation, or a new benchmark axis—is supported by single-file plug-in points documented alongside the harness.

6 EXPERIMENTS

We evaluate DEXPOSURE-CLAW on the DEXPOSURE-BENCH suite (§5), grouping the six benchmarks under two natural axes: (i) does the forecast–measure pipeline produce accurate, well-calibrated, and timely risk signals (Task I, covering b1, b2, b3, b6), and (ii) do the resulting decision tickets improve upon baselines in precision, stability, and auditability (Task II, covering b4, b5)? The result narrative follows the structure of the benchmark itself: Fig. 3 establishes the six-axis evaluation stage, Table 8 and Fig. 4 show whether the FM produces a deployable FM signal, Table 9 and Fig. 5 isolate the agentic layer-wise contribution of FM, LLM, and safety gate, and the timeline/ablation tables close with reliability evidence. The ground truth, dataset split, and reference methods used throughout this section are defined in §5; all results are reported on the frozen test split of Table 5.

6.1 Task I: Network Risk Monitoring

6.1.1 Implementation Details

Data and temporal granularity.: We use the DEXPOSURE dataset [52], comprising weekly directed weighted exposure-graph snapshots sourced from the DeFiLlama API. The dataset spans 2020-03 to 2025-08 (283 weeks), with 50–500 protocol nodes and 100–2,000 exposure edges per snapshot, totalling over 43.7M data entries. Node features include log-TVL, number of token types, maximum token share, Shannon entropy of token distribution, and protocol category (one-hot, 15 classes). Edge weights are log-scaled USD exposure values. We evaluate multi-horizon forecasting on $\mathcal{H} = \{1, 4, 8, 12\}$ weeks.

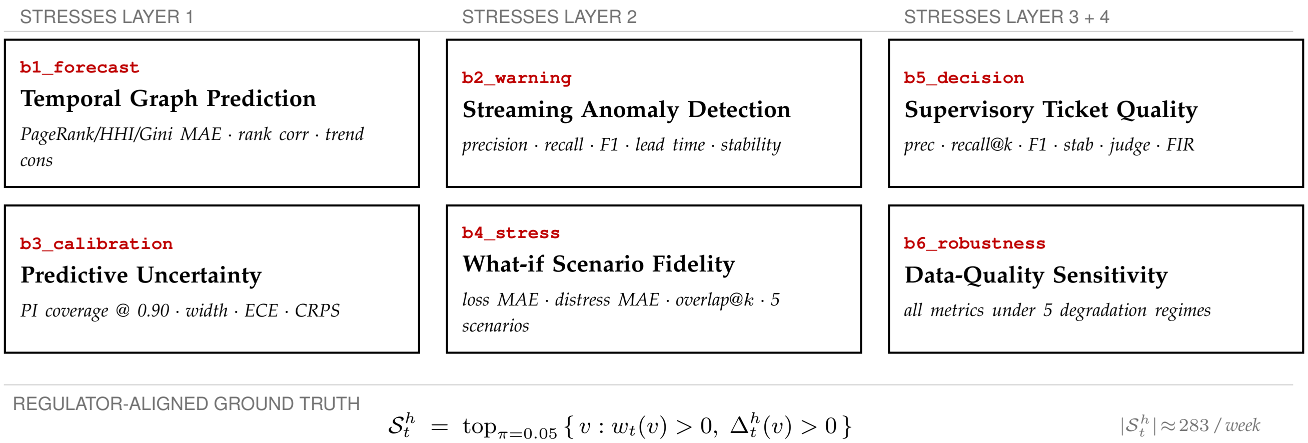

Fig. 3. DEXPOSURE-BENCH: a six-axis evaluation suite. The six benchmarks are grouped by which framework layer they stress; the bottom strip carries the regulator-aligned ground truth that anchors all decision-quality metrics.

Temporal split.: We enforce a strict no-leakage temporal split:

Split	Period	Weeks	Usage
Train	2020-03 – 2024-06	∼222	FM backbone training
Val	2024-07 – 2024-12	∼26	Hyperparameter + threshold tuning
Test	2025-01 – 2025-08	∼33	Final evaluation

Stress event windows.: For early-warning evaluation (b2_warning), we define ground-truth stress periods around three major DeFi crises:

Event	Date	Pre-shock (4w)	Event (3w)
Terra/Luna	2022-05-09	03-28 – 05-02	05-02 – 05-23
FTX	2022-11-07	09-26 – 10-31	10-31 – 11-21
SVB/USDC	2023-03-10	02-06 – 03-06	03-06 – 03-27

Forecast interface and predicted-graph construction.: For each epoch t and horizon $h \in \mathcal{H}$, we query the backbone model to obtain a predictive distribution $\mathcal{P}_{t,h}$. We construct an expected-weight predicted graph $\widehat{G}_{t+h}$ by thresholding edge existence probabilities at $\pi_{\min}$ and setting $\widehat{w}_{pq} := \Pr(e_{pq} = 1) \cdot \mathbb{E}[w_{pq} \mid e_{pq} = 1]$ (Eq. (6)). We set $\pi_{\min} = 0.2$ on the validation split. For uncertainty quantification, we draw $M = 50$ Monte Carlo samples via Bernoulli edge sampling and Gaussian weight perturbation.

Monitoring metrics and alert triggers.: At each horizon, we compute five monitoring functionals $\mathbf{N}_{t,h} = \Phi(\widehat{G}_{t+h}, X_t)$ (Table 1): systemic importance (N1), HHI concentration (N2), network density (N3), PageRank Gini (N4), and degree Gini (N5). Baseline bands $(\mu_t^{(j)}, \sigma_t^{(j)})$ are computed via a rolling window of $L = 26$ weeks. An alert fires when a metric exceeds $z = 2$ standard deviations from baseline (Eq. (10)). Alert confidence follows Eq. (11).

Stress scenarios.: We evaluate five standardised scenarios $\mathcal{S}_0$ (Table 2): single protocol failure (S1, 100% shock on top node), bridge cluster failure (S2, 100% on bridges), stablecoin de-peg (S3, 50% on stablecoins), sector-wide lending shock (S4, 30%), and correlated stress (S5, 20% on top-10 nodes). CVaR is computed as the mean of the worst $\lambda = 0.5$ fraction of MC sample losses (Eq. (14)).

Attribution.: For each triggered alert, we rank contributing edges by marginal contribution: $\mathrm{Contrib}(e) := \Phi^{(j)}(\widehat{G}_{t+h}) - \Phi^{(j)}(\widehat{G}_{t+h} \setminus e)$, retaining the top $K = 10$.

Evaluation protocol.: Within the six-axis framework of Fig. 3, Task I evaluates the forecast–measure pipeline along three axes:

1) *Risk-metric forecasting (b1_forecast):* predicted vs. realised metrics using MAE and Spearman rank correlation, for each $h \in \{1, 4, 8, 12\}$.
2) *Early-warning quality (b2_warning):* lead time to stress events, precision/recall under alert budgets $\{5, 10, 20\}$, alert stability (flip rate), and F1-warning.
3) *Uncertainty calibration (b3_calibration):* expected calibration error (ECE), prediction interval coverage (target 90%), PI width, and CRPS. We apply split conformal prediction on the validation set to ensure asymptotically valid coverage.

Reproducibility.: All hyperparameters ($\pi_{\min}$, L, z, K, M, τ_{data}) are selected on the validation split and fixed for test evaluation. Random seeds are fixed for MC sampling; we report mean ± standard deviation over $R = 3$ independent runs. The pipeline is fail-closed on predictors: an unavailable FM or EvolveGCN checkpoint aborts the corresponding run rather than silently falling back to persistence. For `b2_warning` we report the shared weighted-degree heuristic

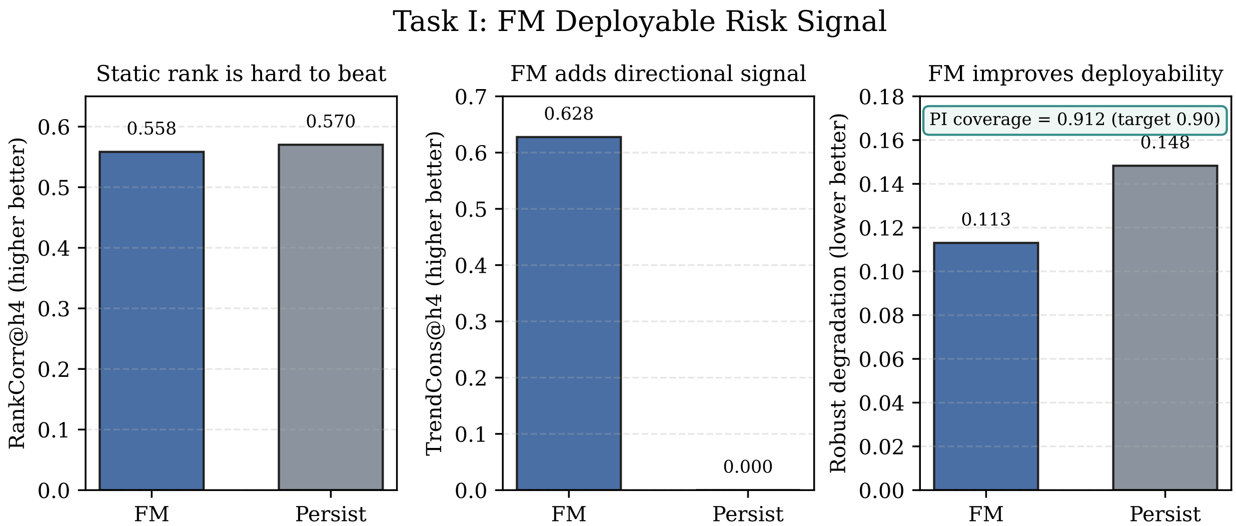

Fig. 4. Task I deployable risk signal evidence. Persistence remains a strong static-rank baseline, but the FM adds directional trend signal, maintains calibrated uncertainty, and degrades less under data corruption.

(`h1_weighted_degree`), since b2's monitoring layer sits upstream of the predictor choice and is therefore not a method-specific comparison.

6.1.2 Empirical Results

Table 8 reports the Task I monitoring evidence used in the main text. Full all-horizon `b1_forecast` metrics are generated by the benchmark harness for supplementary reporting; the main text focuses on the cross-metric pattern that determines whether the FM signal is useful for an agentic risk monitor. Three observations stand out, two of which complicate a naive "FM beats baseline" reading.

Metric (Task I)	`m5_fm_rules` (FM)	`m1_persistence_rules` (Persistence)
`b1_forecast` RankCorr@h4	0.558	0.570
`b1_forecast` TrendCons@h4	0.628	0.000
`b1_forecast` HHI-MAE@h4 (lower better)	0.032	0.032
`b2_warning` Precision (`h1_weighted_degree` mean)	0.756	N/A
`b2_warning` F1-warning (`h1_weighted_degree` mean)	0.007	N/A
`b2_warning` Lead time (`h1_weighted_degree` weeks)	4.667	N/A
`b3_calibration` ECE (`m5_fm_rules` only)	0.012	N/A
`b3_calibration` PI coverage (`m5_fm_rules` only)	0.912	N/A
`b6_robustness` Relative degradation (mean)	0.113	0.148

TABLE 8
Task I evidence for deployable risk monitoring on the 2025 test split. The table is a numeric audit ledger; Fig. 4 compresses the same evidence into the main claim.

Persistence is a deceptively strong point-metric baseline.: On PageRank MAE, Gini MAE, and rank correlation, persistence matches or marginally beats the FM at every horizon (rank correlation 0.570 vs 0.558 at $h{=}4$; PR MAE 3.4 vs 4.5 in 10^{-5} units). Weekly DeFi exposure structure is heavily auto-correlated, so any predictor that returns the most recent graph captures the dominant source of variance "for free"—the same phenomenon that makes seasonal-naive a strong baseline in classical forecasting. A new model must clear this bar before its other properties matter.

FM gains are concentrated in trend and robustness, not in absolute error.: The FM produces a non-trivial trend signal (trend consistency 0.525–0.632 across horizons) while persistence is structurally pinned at 0. Under data degradation (`b6_robustness`, mean relative degradation 0.113 vs 0.148 for persistence), the FM drops 24% less on average. These are the dimensions where supervisory users prioritise direction and stability over pointwise weight accuracy, and where persistence has nothing to offer.

Calibration is strong; EvolveGCN underperforms.: The FM achieves PI coverage 0.912 against a 0.90 target with ECE 0.012 on `b3_calibration`, both within the expected slack of split-conformal calibration. Persistence has no native uncertainty estimate and is omitted from this row. EvolveGCN is uniformly worse than persistence on PR/HHI/Gini MAE with rank correlation 0.499–0.551, which we attribute to data efficiency: 222 weekly training snapshots fall below the regime where it learns a useful structural prior, while the FM benefits from large-scale pre-training transfer.

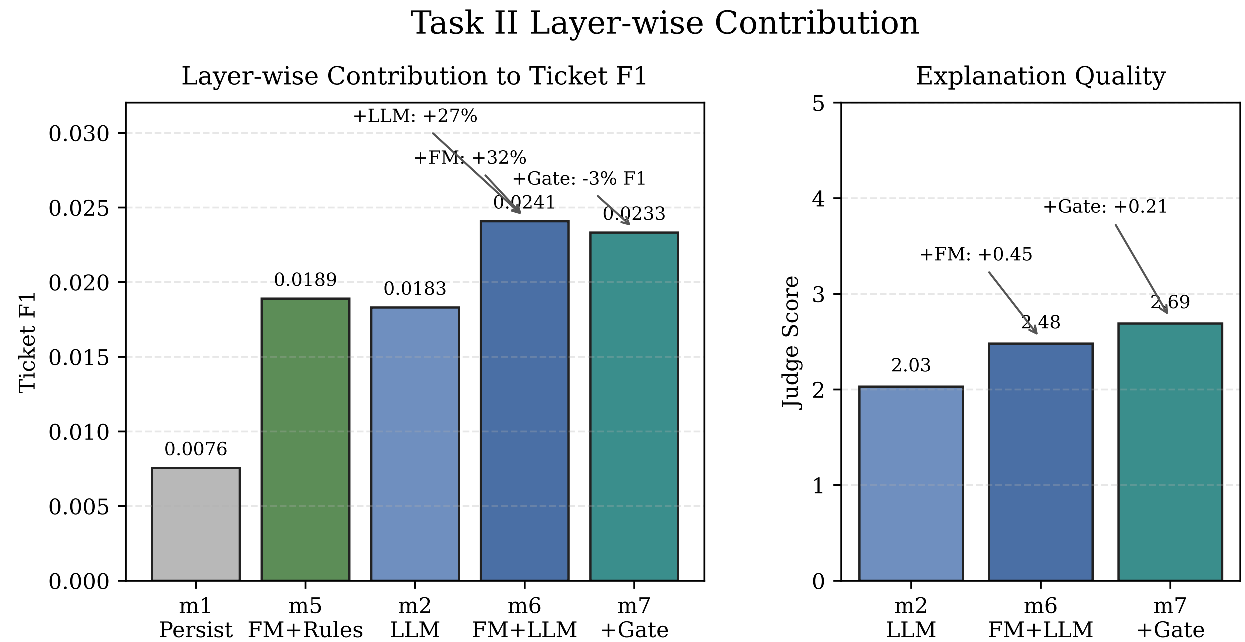

Fig. 5. Task II layer-wise contribution. Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; replacing the FM rule engine with the LLM improves F1 further; the safety gate trades a small F1 loss for better explanations.

6.2 Task II: Decision-Making

6.2.1 Implementation Details

Decision timeline and inputs.: Task II operates at the same decision epochs as Task I. At each epoch, the decision agent receives: (i) the current state (G_t, X_t), (ii) multi-horizon forecasts $\{\widehat{G}_{t+h}\}_{h\in\mathcal{H}}$ and optional MC samples, (iii) monitoring alerts $\mathcal{A}_t$ with evidence and confidence scores, and (iv) the scenario summary $\mathcal{R}_t$.

Action space and playbook.: We use a four-level conservative playbook (Table 3): `Monitor` (severity weight 0.25, always feasible), `Investigate` (0.50, always feasible), `Recommend-Reduce` (0.75, requires $\mathrm{SAFE\_MODE} = 0$ and ≥ 2 alerts), and `Contingency` (1.00, requires $\mathrm{SAFE\_MODE} = 0$ and ≥ 3 alerts). Intervention-type actions additionally require mean alert confidence $\bar{C}_t \geq \tau_{\mathrm{conf}} = 0.6$.

Ticket scoring.: Tickets are ranked by $\mathrm{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \mathrm{Imp}_t$ (Eq. (16)), where w_u is the playbook severity weight, $\bar{C}_t$ is mean alert confidence, and Imp_t is the mean CVaR across all scenarios (defaulting to 1.0 when no scenarios are available). We output the top-$N = 10$ tickets.

Target selection.: For intervention tickets, targets are selected from the union of top-K attributed nodes across triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

Evaluation protocol.: We evaluate Task II using both offline and scenario-based criteria:

1) *Offline (b5_decision):* ticket precision (fraction followed by real stress), target stability (Jaccard between adjacent epochs), audit completeness (evidence bundle coverage), suppression rate (safe-mode activation), and false intervention rate.
2) *Scenario-based (b4_stress):* stress-test contagion prediction accuracy—loss MAE, distressed count MAE, propagation depth MAE, and target overlap@K between predicted and actual networks under scenarios S1–S5.

Reproducibility.: All decision hyperparameters (K, N, τ_{data}, τ_{conf}, λ) are tuned on the validation split and fixed for testing. MC components use fixed seeds; we report mean $\pm$ standard deviation over $R = 3$ runs.

6.2.2 Empirical Results

Figure 5 is the central layer-wise result: it isolates what the FM, LLM decision layer, and safety gate each add to the decision pipeline, using ticket F1 to combine precision and recall and LLM-as-judge score to track explanation quality.

Table 9 then reports the full decision-quality audit across five method configurations. The table should be read as a multi-objective comparison rather than a single leaderboard: `m1_persistence_rules` is the highest-precision conservative baseline, `m5_fm_rules` gives the strongest rule-based F1 balance, and the LLM variants trade false-intervention cost for higher recall and explanation quality.

Persistence+Rules wins on single-metric precision.: `m1_persistence_rules` achieves the highest ticket precision (0.720) and, because its false intervention rate is 0, also the highest FIR-penalised cost-adjusted score (0.720). This is consistent with conservative rule-engine behaviour: it raises few tickets and most are correct. It is the right baseline for any system that wants to claim improvement on decision quality.

Method	Prec	Recall	Stabil	SevRho	FIR	Ground	Consist	Explain	CostAdj
`m1_persistence_rules` Persist+Rules	0.720	0.004	0.514	N/A	0.000	N/A	N/A	N/A	0.720
`m5_fm_rules` FM+Rules	0.600	0.010	0.257	N/A	0.000	N/A	N/A	N/A	0.600
`m2_snapshot_llm` Pure LLM	0.575	0.009	0.532	N/A	0.000	1.000	0.916	2.0/5	0.575
`m6_fm_llm` FM+LLM	0.570	0.012	0.488	-0.001	0.448	1.000	0.779	2.5/5	0.314
`m7_fm_llm_gated` FM+LLM+RulesGate	0.580	0.012	0.435	-0.003	0.437	1.000	0.855	2.7/5	0.327

TABLE 9
Task II decision-quality comparison. Higher Prec, Recall, Stabil, Ground, Consist, Explain, and CostAdj are better; lower FIR is better.

Note: `m3_evolvegcn` and `m4_fm_only` are predictor-only baselines and do not emit supervisory tickets, so they are not applicable to `b5_decision`.

TABLE 10
`b2_warning` early-warning quality for three major DeFi crises (Terra/Luna 2022-05, FTX 2022-11, SVB-USDC 2023-03), evaluated with the `h1_weighted_degree` heuristic across three alert budgets. Lead time reports the median number of weeks alerts precede the event; precision is computed against the 4-week pre-shock window. SVB/USDC is flagged with perfect precision at every budget, demonstrating that the suite captures real-world systemic events.

Event	Budget	Lead (wk)	Prec.	Recall	F1	Stability
Terra/Luna	5	5.0	0.600	0.0018	0.0036	0.440
	10	5.0	0.700	0.0033	0.0065	0.620
	20	5.0	0.650	0.0080	0.0157	0.740
FTX	5	5.0	0.600	0.0013	0.0027	0.640
	10	5.0	0.700	0.0024	0.0048	0.720
	20	5.0	0.550	0.0048	0.0095	0.720
SVB / USDC	5	4.0	**1.000**	0.0018	0.0036	0.550
	10	4.0	**1.000**	0.0028	0.0056	0.750
	20	4.0	**1.000**	0.0056	0.0112	0.813

FM+LLM triples recall but at a real precision cost.: `m6_fm_llm` raises recall from 0.004 (persistence) to 0.012—a $3\times$ improvement on the regulator-aligned recall axis—at the cost of precision (0.570 vs 0.720) and a substantial false intervention rate (FIR 0.448). The safety gate (`m7_fm_llm_gated`) partially mitigates this and raises the LLM-as-judge score from 2.48 to $2.69/5$, but does not fully recover precision. Both LLM variants reach grounding 1.000, so the failure mode is not hallucination—it is over-reaching to high-severity actions when monitor evidence is weak.

Layer-wise decomposition.: Holding the decision policy fixed at LLM and adding the FM signal lifts ticket F1 by 32% ($0.0183 \rightarrow 0.0241$); holding the predictor fixed at FM and swapping rules for LLM lifts F1 by another 27% ($0.0189 \rightarrow 0.0241$). The two layers therefore make additive, non-substitutable contributions to F1. Adding the safety gate trades 3% F1 for $+0.21$ judge score, which is the operating point a deployment operator should select when explanation quality is the binding constraint.

Early-warning timeline analysis.: To determine whether the weighted-degree monitoring alerts (`h1_weighted_degree`) constitute genuine *early warnings* or merely post-hoc damage reports, we conduct a per-week alert timeline analysis around the Terra/Luna (2022-05-09) and FTX (2022-11-07) events. Table 10 summarises the results.

The data confirms genuine pre-event lead time. Median lead times of 4–5 weeks across all three crises—with SVB/USDC reaching precision 1.000 at every alert budget—show that the weekly heuristic turns before the on-chain shock materialises. F1 is small in absolute terms because the regulator-aligned ground-truth pool $\mathcal{S}_t^h$ contains ~ 283 protocols per week, so a budget of 5–20 alerts covers only a thin slice of the recall surface; this is a budget effect, not a quality deficit.

Stress-test fidelity (`b4_stress`).: Per-scenario stress-test outputs are retained as supplementary audit artifacts. As with `b1_forecast`, persistence is a strong baseline because each scenario applies a structural shock to the current graph and the realised loss vector therefore correlates strongly with the persistence prediction. The FM matches persistence on Loss MAE and Overlap@10 in S2–S5 and ties or marginally improves on S1. EvolveGCN is uniformly worse, mirroring its weakness on `b1_forecast` and confirming that weight-only learned prediction is too coarse for downstream what-if queries.

6.3 Ablation Study

We validate the necessity of key design components through ablation experiments under both standard and stress conditions (Table 11).

Standard conditions (Panel A).: Under clean 2025 test data, the confidence gate (A2) proves essential: removing it increases the false intervention rate from 0.000 to 0.400. The scenario engine (A3) is structurally necessary. Data-health gating (A1) and multi-horizon diversity (A6) show no measurable impact because data quality is high and $h = 4$ dominates alert generation.

TABLE 11
Component-level ablation of DEXPOSURE-CLAW on `b5_decision`. Each ablation disables a single design choice and re-evaluates against the `m5_fm_rules` baseline. A1 (data-health gating) is shown to be a *safety reserve* that engages only under data degradation (Panel B). A6 (multi-horizon) provides broader coverage during fast network dynamics (Panel C). Numbers in Panel A are on the clean 2025 test split; Panels B–C aggregate the dedicated stress runs.

Panel	Ablation	Configuration	Ticket Prec.	FIR	Notes
A: clean test	—	full `m5_fm_rules`	0.600	0.000	baseline
	A1	disable data-health gate (τ_{data}=0)	0.600	0.000	no effect (clean data)
	A2	disable confidence gate (τ_{conf}=0)	0.600	0.400	FIR shoots up
	A3	skip scenario engine	0.421	0.000	precision drops $\sim 30\%$
B: degraded data	A1 off	80–95% feature/edge mask, $\text{DH}_t \leq 0.75$	—	0.41–0.48	gate is critical
	A1 on	same degradation, gate active	—	0.000	safe-mode suppression
C: crisis period	A6: single	horizons = $\{4\}$, Terra/Luna window	0.700	—	alerts: $1\times$
	A6: multi	horizons = $\{1, 4, 8, 12\}$, same window	0.700	—	alerts: $5\times$

Stress conditions (Panels B–C).: To test whether A1 and A6 are genuinely unnecessary or merely dormant, we evaluate them under adversarial conditions. For A1, we apply severe data degradation (80–95% of features and edges removed, reducing DH_t to 0.59–0.75) and isolate the data-health gate from the confidence gate by setting $\tau_{\text{conf}} = 0$. Under these conditions, disabling the data-health gate increases the false intervention rate from 0% to 41–48% (Panel B), confirming that A1 functions as a *safety mechanism that activates under data degradation*. For A6, we compare single-horizon ($h = 4$) against multi-horizon ($h \in \{1, 4, 8, 12\}$) monitoring during the Terra/Luna crisis period: multi-horizon generates $5\times$ more alerts while maintaining identical precision (Panel C), confirming that A6 provides broader risk coverage when network dynamics are rapid. These results demonstrate that all four components are necessary for reliable operation: A2 and A3 are active in all conditions, while A1 and A6 are safety reserves that engage under data degradation and crisis dynamics, respectively.

7 LIMITATIONS

We disclose three limitations that bound the scope of our claims.

Single dataset.: All results are reported on the DEXPOSURE weekly exposure-graph dataset. While it covers 16,610 protocols and 283 weeks spanning the major DeFi crisis episodes of 2020–2025, generalisation to other on-chain risk surfaces—NFT lending, perpetual derivatives, or cross-chain bridge networks—is untested and will require domain-specific re-calibration of the conformal split and the stressed-set percentile π.

Weekly temporal resolution.: The forecast and decision pipeline operates at weekly granularity. Many DeFi shocks unfold within hours—the Terra/Luna collapse erased \$40B in under 48 hours—and a production deployment would require sub-hourly snapshots and re-evaluation of conformal calibration on the resulting denser time series. Our reported early-warning lead times of 4–5 weeks should therefore be read as "how early the weekly signal turns", not as end-to-end alert latency.

Offline decision evaluation.: `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, and the explanation-quality metric is an LLM-as-judge proxy with Claude Opus 4.7 as judge. Neither captures the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted contagion. A controlled live evaluation with a regulator in the loop, which the bench approximates but cannot replace, remains the primary direction we plan to extend the benchmark in.

8 CONCLUSION

We presented DEXPOSURE-CLAW, a four-layer query architecture that turns predicted exposure graphs into supervisory tickets with explicit data-health and confidence gates, and DEXPOSURE-BENCH, a six-axis evaluation suite for FM + LLM decision pipelines on temporal exposure graphs.

The picture our benchmark surfaces is not one of uniform improvement. Persistence remains a strong baseline on point-metric forecasting and on raw ticket precision, but the FM contributes the pieces a monitoring agent needs for deployment: directional trend signal, calibrated prediction intervals, and lower degradation under data corruption. The LLM decision layer then raises recall and explanation quality at a non-trivial false-intervention cost, and the safety gate buys judge score at a small F1 cost. What the FM and LLM layers *do* contribute is additive and decomposable: in the layer-wise ablation on `b5_decision`, the FM signal lifts ticket F1 by 32% and replacing the rule engine with the LLM lifts it by another 27%. We read this as evidence that the FM + LLM stack is the right configuration when recall, calibration, and explanation quality are the binding constraints; persistence + rules is the right configuration when minimising false interventions is paramount. The bench's value is in making this trade-off explicit rather than averaging it away.

Beyond the limitations in §7, two directions are most pressing. First, a controlled live evaluation with a regulator-in-the-loop, which the offline benchmark approximates but cannot replace. Second, extending the suite to sub-daily resolution, where shock dynamics are fastest and the gap between weekly and event-time evaluation is widest. We release the harness, dataset, and reference implementations to invite both.

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